# SOP-ENGINEERING CHANGE REQUEST MANAGEMENT

[CONFIDENTIAL]
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## CHANGE HISTORY

This section describes the details of changes that have resulted in the current document.

| SL. No. | Document Version No. | Change Description | Reason For change | Date Prepared By | Date Reviewed By | Date Approved By |
|---|---|---|---|---|---|---|
| 01 | 0.1 | Draft | Implementation of the Integrated Management System | Mallikarjuna BN 30-Oct-23 | | |
| 02 | Issue B Version 1.0 | Review Comments | | | Vijay Kumar GS 30-Oct-23 | Vivek SK 02-Nov-23 |
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| 04 | Issue B Version 1.2 | Process improvements | Enhancements to all sections | Siva Prasad P 10-Jan-23 | Anitha N 9-Feb-24 | Santosh V 9-Feb-24 |
| 05 | Issue B Version 1.3 | Document formatting done as per document guidelines | Updating document as per document guidelines | Shareef Shaik 31-Dec-24 | Anitha N 31-Dec-24 | Santosh V 31-dec-24 |
| 6 | Issue C Version 1.0 | Review of document | New version update of IMS | Anitha N 01-08-2025 | Santosh V 08-Aug-25 | Sanmangala MS 14-08-2025 |

# 1 Purpose

The purpose of an Engineering change request management (ECRM) is to facilitate a controlled and documented process for introducing modifications, updates, or improvements to engineering designs, products, or processes within an organization with primary objectives like change management, continuous Improvement, risk mitigation, regulatory compliance, quality assurance, cross-functional collaboration**.**

# 2 Scope

The scope of an Engineering change request management (ECRM) typically covers a range of activities related to proposing, reviewing, approving, and implementing changes in engineering designs, products, or processes, key elements within the scope of an ECRM process like design changes, process changes, material changes, notification and communication, timely implementation, customer impact, auditability, and documentation.

# 3 Abbreviations

a. ECN - Engineering change notice

b. PM - Project manager

c. SVN - Source version control

d. ECO- Engineering change order

e. ECR- Engineering change request

f. RFQ - Request for quotation

g. CCB - Change control board

h. QA - Quality assurance

i. QMS - Quality management system

j. APL - Approved supplier list

k. NDA - Non-disclosure agreement

l. IGI - Incoming goods inspection

m. PCB - Printed circuit board

# 4 Terms and Definitions

a. ECR: Formal document initiated by stakeholders within to suggest modifications, improvements, or corrections to a product, system, or process, and suggested ways to address or fix the problem, what items would have to change to fix it, and

signoffs to say the suggestions are approved and to move forward with the ECO process.

b. ECO: Formal document that authorizes and implements approved changes to a product, system, or process. It details the modifications required, specifies the affected components, and provides instructions for implementation.

c. ECN: Documents notify the appropriate or interested parties when the ECO change(s) are complete.

d. Product change: Product change refers to any modification, adjustment, or alteration made to a product, whether it be a physical item, software application, or service. These changes can be initiated for several reasons, including improvements in functionality, design enhancements, bug fixes, compliance with standards, cost reduction, or adaptation to changing market conditions.

e. Regulatory change: the regulatory bodies change their regulations in response to different observations, complaints, and new research. When they change the regulatory requirements, it becomes necessary for the accredited manufacturers to update their processes and systems according to the revised requirements.

f. Cross-Functional Collaboration: Encompasses collaboration among different departments, including engineering, production, quality assurance, and documentation, to ensure coordinated efforts in implementing changes.

g. Documents change: An engineering change order is also initiated when any changes in the process documentation or a product are made, documentation change occurs in response to a change in product specification, regulatory requirements, or a significant structural change in a company.

# 5 Responsibilities

a. The responsibilities associated with engineering change requests (ECRs) typically involve a collaborative effort among various stakeholders. Here are key responsibilities related to ECRs:

b. Initiator: Propose the need for a change in design, specifications, or any other engineering aspect with relevant documentation or supporting information

c. Review Board or Change Control Board (CCB): Approve or reject the ECR based on the evaluation of the proposed changes in a systematic manner to determine its impact and feasibility and to assess the technical, financial, and operational implications of the proposed changes.

d. Engineering Team: Implement the approved changes in the design or engineering specifications and ensure that the changes align with quality standards and regulatory requirements.

e. Configuration Manager: Update relevant documentation to reflect the approved changes and to Maintain version control of updated documents for revise manuals, procedures, or any other documentation affected by the engineering change.

f. Quality Assurance (QA): Ensure that the changes comply with quality standards and do not introduce new risks or defects and perform inspections or tests related to the changed components.

g. Project Management: Project managers track the progress of the ECR process and ensure that it aligns with project timelines, monitor the implementation of changes, and communicate status updates to relevant stakeholders.

h. Verification and Validation: Engineering team is responsible for the implemented changes achieve the intended objectives by verification to ensure that the changes were implemented as specified and validate that the changes meet the initial objectives and do not introduce unintended consequences.

i. Documentation of Changes: Project teams maintain a comprehensive record of all changes for future reference and archive relevant records, ensuring traceability and auditability.

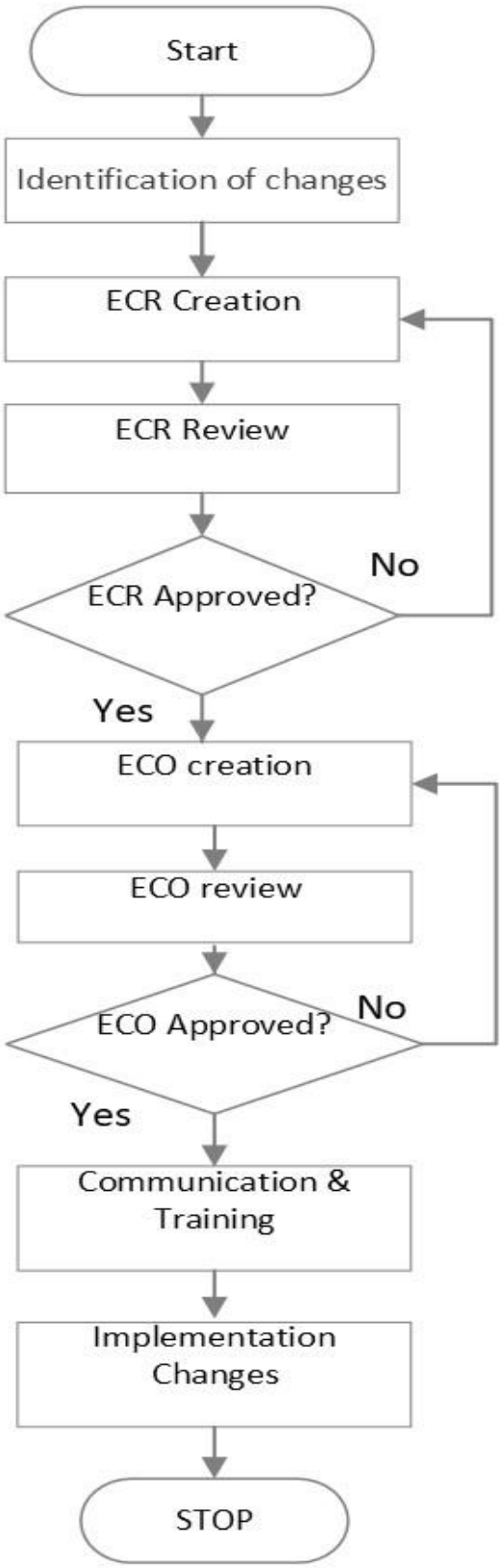
# 6 RACI

| Process | Responsible | Accountable | Consulted | Informed |
|---|---|---|---|---|
| Identification of Changes | Engineering Team | Engineering Manager | Cross functional team | Delivery Manager, Customer |
| ECR Creation | Engineering Team & Project manager | Engineering Manager | Cross functional team | Delivery Manager |
| ECR Approval | Project Manager | CCB | Delivery Manager | BU Head |
| ECO Creation | Engineering Team | Engineering Manager | Project Manager | Delivery Manager |
| ECO Approval | Project Manager | CCB | Delivery Manager | BU Head |
| Communication & Training | Engineering Team | Engineering Manager | Project Manager | Delivery Manager |
| Implementation | Project Manager | Engineering Team | Delivery Manager | BU Head |

# 7 SIPOC

| Supplier | Input | Process | Output | Customer |
|---|---|---|---|---|
| Engineering Team | Change requests | Identification of Changes | Identified Change requests | Project Manager |
| Engineering team | Identified Change requests | ECR Creation | Documented in ECR Form | CCB |
| Project Manager | ECR Form | ECR Approval | Approved ECR | Engineering team |
| Engineering team | Approved ECR | ECO Creation | Detailed ECO | CCB |
| Project Manager | Detailed ECO | ECO Approval | Approved ECO | Engineering Team |
| Engineering Team | Approved ECO | Communication and Training | Communication and trained on required skills to engineering team | Engineering Team |
| Project Manager | Communication of ECO and trained engineering team | Implementation | Implemented CR | Relevant Stakeholders |

# 8 Overall Process Flow Chart



# 9 Process description

a. Engineering change request management must be authorized by the change control board. ECRM shall go through different steps to effectively implement the change.

b. Below mentioned steps shall be followed in an engineering change request management process:

- Identification of changes.
- Investigation of the need for change and justification.
- Engineering change request (ECR) creation.
- Engineering change request (ECR) review.
- Engineering change order (ECO) creation.
- Engineering change order (ECO) review.
- Engineering change order (ECO) approval.
- Communication and training.
- Implementation.

c. The ECRM process begins with a requester who identifies a need and advantage after the implementation of change compared to the existing design/setup in the engineering domain.

d. The requester completes an ECR form, providing details such as change description, rationale, affected components, and any supporting documentation.

e. Project manager and technical team performs an initial review of the change, its impact, and the requirements to ensure that the ECR form is complete and aligns with the established criteria.

f. The engineering team evaluates the technical feasibility of the proposed change, considering factors like design, compatibility, and performance.

g. Relevant stakeholders, including engineering, quality, and production, participate in a cross-functional review to assess the proposed change's impact and conduct a risk analysis to identify potential risks associated with the change and develop mitigation strategies and evaluate the cost implications of the change, considering material costs, labor, and potential impact on existing inventory before proceeding toward the final implementation stage.

h. The ECR moves through an approval workflow, involving relevant stakeholders and CCB.

i. After approval of ECR, the engineering team creates a detailed engineering change Order (ECO).

j. ECO is the subsequent stage that follows the approval of an ECR to implement the approved changes with detailed instructions on how the change should be implemented, including technical specifications how to modify the product or process, timelines, and resource requirements, ensuring minimal disruption to ongoing operations.

k. Engineering change order (ECO) is shared with change control board (CCB) for final review.

l. The change control board (CCB) thoroughly studies various aspects of the process, such as the necessity of change, post-change scenario, and overall impact on the performance of the product and company.

m. Each member analyzes from their expertise, after consensus, an engineering change order (ECO) is approved. Otherwise, the change is not implemented and is revert back for revision.

n. IF ECO is approved, the project manager will inform all the concerned personnel about the approval of the change, to start implementing the change as per the approved plan.

o. Training is conducted for all the concerned personnel involved in the implementation of change, the main purpose of conducting training is to follow the approved plan, specifications, and requirements. the training also ensures the implementation of change according to the desired deliverables.

p. The implementation stage is the practical form to Implement the approved changes according to the detailed plan like resource allocation, task implementation, and monitoring performance.

q. The engineering team will conduct verification and validation activities to ensure that the implemented changes meet the intended objectives.

r. Project managers evaluate the effectiveness of the implemented changes, identifying any unforeseen issues or deviations from the plan.

s. The engineering change order (ECO) and the engineering change notice (ECN) are the guiding rules and principles to be followed.

t. Project team capture and document lessons learned from the Engineering Change Request Management (ECRM) process, offering valuable insights for future change management endeavors.

u. Project manager will conclude the ECRM with meticulous documentation, encompassing final approvals, implementation specifics, and findings from post-implementation reviews.

## 10 Information Security requirements

a. Information Security incidents and complaints are logged and closed appropriately.

b. Confidentiality of information and project details is maintained by the team.

c. Archiving and backing up of data as per contractual obligations.

## 11 References

a. ISO 9001 Clauses 6.3.

b. ISO 27001 Clauses 8.1, Controls A.14.

c. ISMS Control Policies.

## 12 Metrics

a. Total ECR/ECO Identified and Implemented.

b. Total ECR/ECO Approved and Rejected.

c. Schedule & Cost impact for implementation ECR/ECO.

## 13 Annexures

NIL

## 14 Forms

a. Engineering Change Note UT/SD/DSP/TMP/07.

b. Change Request Form UT/PS/CMP/TMP/03.